

AIR-GAP SALVAGE

FIELD SERVICE LOG & SUB-CANAL REPAIR NARRATIVE



HARDWARE SYSTEM

MNIME-75 / Edge Mod-C

OPERATING LOCATION

Lukas Sundries, Lower Canal

OPERATIONAL MODE

Zero-Cloud // 100% Uncoupled

SERVICE INCIDENT

Token Window Cascade [Cleared]

OPERATIONAL ABSTRACT

A record of off-grid machine extension servicing in the Lower Basin drainage flats, 2075. Documents the stabilization of an overloaded edge co-processor suffering from recursive context overflow, and the mechanics of local, non-mesh neural inference performed behind the walk-in cold box. Features full field telemetry readouts, bus schemas, and procedural token purges.

AIR-GAP SALVAGE

LOG RECORD // SUB-CANAL UNIT SERVICING // REF: MNIME-75

NON-MESH // SECTOR 4

DOCKET 01.1 // THE FRONT COUNTER AND THE COUNTERWEIGHT

The amber sign above the awning sputtered through a shroud of oily October moisture: *LUKAS GROCERY & SUNDRIES*. Below the fogged glass, sagging composite shelves were lined with vacuum-sealed plantain biscuits, five-kilo tubs of hydrogenated lard, gray marsh rice, and dented tins of mock-mackerel. It was four in the morning along the canal sluices. The low-tide stench of rotting bladderwrack, sump fuel, and oxidized copper rolled off the retaining walls, thick enough to leave a bitter scum on a man's tongue.

The bead curtain rattled with a brittle metallic scrape. A kid slipped through, shoulders hunched forward to guard his left side. He wore an oilcloth windbreaker crusted with caustic road slag, and he didn't glance at the bread bins or the bins of salt-pork. He moved with the rigid, off-balance gait of someone carrying dead weight inside his neck. He walked straight to the meat slicer, pulled a scorched brass token from his wrist sleeve, and pressed it hard against the stainless-steel prep board with a wet, trembling thumb.

SCHEMATIC REF: MNIME-75 / MOD-C

PINOUT SPEC // GROUNDED EDGE BUS



Lukas did not look up from dragging a coarse yellow rag across the brine rail. He had been slicing pickled beef since midnight, his hands smelling of clove oil and curing salts. "Cold water is free at the spigot by the sink," he muttered, voice gravel-dry. "Dry rations in the cedar bin. Back rooms are shut until morning delivery."

"The token's brass-cast and solid, mister," the kid croaked. His left eyelid fluttered with an involuntary, erratic tremor—a rapid firing of ocular nerves triggered by stray electrical bleed. "I'm dropping frames every three seconds. The core is bleeding unquantized tensor branches right into my vestibular canal. I can't tell if the floor is level or if it's drawing an orange vector grid under my boots."

Lukas stopped his hand. He tossed the rag into an enamel bucket and finally inspected the collar of the boy's jacket. Anchored beneath the frayed rim of the collar was a bolted, low-profile bracket—the telltale sub-cranial cradle of an MNIME-75 unit. The Multimodal Neural Interface Machine Extension. In the glass towers up in New Harbor, executives ran these things tethered to high-orbit data constellations, receiving instant algorithmic synthesis through encrypted tightbeams. But down in the canal sloughs, connecting to the cloud meant automated repossession warrants, asset garnishments, and surveillance tags. Down here, you ran local, offline, and cold. Or you didn't run at all.

DOCKET 01.2 // BEHIND THE FREEZER DOOR

"Back of the cold box," Lukas said, jerking his chin toward the heavy galvanized door behind the counter. "Mind the brine barrels. If you knock a seal loose, you pay for the vinegar."

Behind the walk-in's rows of hanging cured pork bellies and wheel-cheeses lay an eight-by-eight alcove lined with recycled acoustic foam and salvaged sheets of copper mesh. It was an improvised Faraday cage that smelled of smoked fat, pickling brine, and pure isopropyl alcohol. A scarred workbench of untreated timber held a flickering cathode monitor, two grounding straps, a variable-temp soldering iron, and a rack of passive heat sinks wired to an old x86 microkernel chassis running on lead batteries.

The boy sat on an upturned wooden crate, teeth chattering from both cold and shock. Lukas clamped a heavy copper alligator lead to the grounding lug on the MNIME bracket. Static discharged with a faint blue snap, sinking stray surface charges into the building's iron water mains.

>> MNIME-DIAG: PORT /dev/ttyEXT0

STATUS: CRITICAL OVERFLOW

CORE_ARCH: 1.5B PARAM // 4-BIT INT GGUF // RUNNING UNGUARDED
VECTOR_STORE: OVERFLOW (4096/4096 TOKENS SATURATED)
AST_ENGINE: RECURSIVE SYNTAX LEAK IN SENSORY CONTEXT PARSER
ERR_CODE: [0x77E4] - UNCLOSED STRING LITERAL IN VISUAL INPUT FEED
TELEMETRY_ROUTE: AIR-GAP VERIFIED // ZERO RADIO EMISSION

Lukas leaned in with a jeweler's loupe and a three-millimeter magnetic driver. He clicked a task lamp on, illuminating the back of the boy's skull. "Who flashed this garbage?" he asked, unseating the unit's thermal shell. A faint puff of cooked dielectric silicone drifted into the freezing air.

"A guy near the scrap barges," the kid muttered, clutching his knees. "Said he fine-tuned it on tactical precinct files. Said it would parse patrol movements and mark checkpoints two blocks ahead. But when I passed through the market arcade, everything locked up. Every face was surrounded by flashing amber boundary boxes that wouldn't resolve. Then raw code started scrolling across my left retina."

"Because the idiot didn't understand local token bounds," Lukas muttered, loosening the second retaining bolt. "He loaded an unquantized attention matrix into a fixed four-thousand token cache without a sliding decay window. Your machine extension isn't just indexing people; it's compiling every neon sign, cracked cobblestone, and exhaust plume into an unpruned abstract syntax tree. Your VRAM filled to capacity, and the parser started writing swap tokens straight across your optic nerve."

The boy shivered, his breathing shallow. "I could feel my thoughts lagging. Like my own words were stuck in a pipeline, waiting for permission to render."

DOCKET 01.3 // COLD IRON AND LOCAL WEIGHTS

Lukas reached for an aerosol canister of dielectric flush. He directed the straw between the microscopic heat fins of the MNIME bracket and blasted three short bursts. Gray canal grit, oxidized solder flux, and dried sweat washed out, spattering against a catch rag on the bench. The boy flinched at the sudden cryogenic bite, but Lukas pressed a firm hand onto his shoulder to keep him anchored.

Lukas wasn't a surgeon, and he certainly wasn't an academic neural theorist. Before the regional utility syndicates collapsed, he had spent twenty years maintaining real-time diagnostic parsers for industrial gantry cranes down in the shipping basins. He understood that whether code was balancing sixty tons of automated container cranes or translating local acoustic vectors into sensory awareness, the physics remained identical: if the buffer leaks, the machine will tear itself apart.

Subsystem	Incoming State	Bench Intervention	Cleared Spec
Vector RAG Cache	100% Locked (Leak)	Hard Ring Buffer Re-partition	256 MB Clean Ring
AST Parsing Depth	Level 18 (Unpruned)	Depth Capped to 6 Levels Max	Real-Time Prune Active
Thermal Shunt	78°C Thermal Throttle	Liquid Bismuth Paste Applied	34°C Under Heavy Load
Mesh Telemetry Radio	Physical Hard-Severed	Verified Solder Cut-Bridge	Zero-RF Emissive

With steady, calloused fingers, Lukas plugged an insulated ribbon cable from his terminal into the unit's serial debug port. The green phosphor CRT screen flickered, painting lime-colored stripes across the cured pork bellies behind him. He loaded a custom micro-pipeline: six hundred lines of bare C compiled without network libraries or remote runtime hooks.

"Focus on the reticle in the middle of the monitor," Lukas commanded quietly. "Don't blink, and don't swallow. I'm severing the unpruned vector nodes and purging the KV cache. It's going to feel like somebody yanked a fistful of dry gravel out of the back of your brain."

He struck the mechanical enter key. The terminal rattled as hexadecimal register dumps flooded the screen: PURGE_KV_STACK... AST_RESET_DEPTH=6... FLUSH_OK. The boy let out a harsh, choking breath. His spine snapped straight, then collapsed forward against the timber workbench, his shoulders trembling violently as the cognitive backpressure dissolved.

The fan in the diagnostic rig wound down to a gentle whisper, its green glow illuminating the quiet dampness of the room.

DOCKET 01.4 // THE WEIGHT OF FREE SAND

For nearly a minute, neither of them moved. The only sound in the alcove was the steady, rhythmic chug of the freezer compressor and the drip-drip of thaw water draining into the galvanized bucket beneath the lard racks. The boy sat upright by slow degrees, blinking away residual phosphor burns. He raised both hands, staring at his palms as though seeing them for the first time in days.

"The orange boxes," the kid whispered, his voice steady now. "They're gone. The street signs outside aren't trying to parse into text trees anymore. It's just... quiet."

"Your context buffer was clamped down to local boundaries," Lukas said, picking up a tube of thermal bismuth paste and spreading a thin gray film across the die before remounting the anodized heat shroud. "You have four thousand tokens to work with. That is plenty to parse room layout, spot armed movement in an alley, and read warnings off a warehouse gate. It is not enough to simulate the whole damned city. Stop trying to compile the world. Let it exist outside your head."

LUKAS GROCERY & PROVISIONS // SERVICE SLIP

REPAIRS: Flushed VRAM Ring Buffer
FIRMWARE: Stencil v3.12 (Local-Only)
STATUS: Air-Gap Intact / Zero RF

REGISTRY: #075-OK

LABOR: 40 Basin Scrip
COMMODITY: 2 Tins Mock-Mackerel
PAID IN FULL

Lukas tightened the four corner bolts in an alternating cross-pattern until the rubber gasket compressed flat against the boy's skin. He reached over, detached the grounding alligator clip, and coiled the braided lead around a copper hook on the workbench.

The kid reached into his oilcloth pocket and pulled out forty crumpled notes of canal scrip, smoothing them against the rough grain of the table alongside two sealed tins of mock-mackerel. Lukas swept the bills into a dented coffee tin without counting them and nudged the fish toward his pantry shelf.

"Take the lower sluice path home," Lukas instructed, pushing the heavy freezer door open to let the warm, salt-scented dawn air roll back in. "The harbor watch brought new scanner dogs to the bridge. But with that radio bridge physically sheared, you're just another shadow in the basin. Keep your head down, stay off the mesh, and eat something that wasn't synthesized in a vat."

The kid pulled his hood over the sealed hardware, nodded once in silence, and disappeared through the grocery aisles. The brass bead curtain chimed softly, then settled. Lukas walked back behind the front counter, picked up his yellow brine rag, and began wiping the salt residue from the glass case as the rising sun turned the canal fog into a blanket of dull, cold pewter.